SmartSafe Register and Bit Configuration Reference

ATmega32A Master, ATmega32A Slave, and 25LC040A EEPROM

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Purpose and Scope

This document is a register-centered companion to the complete SmartSafe project report. It identifies every ATmega32A register directly configured, read, cleared, or temporarily saved by the Master and Slave firmware. For each register, the bit positions, hardware meaning, selected value, and project-level reason are stated. Registers that share an address, flags that are cleared by writing a logic one, and timed write sequences are highlighted because these details are common sources of embedded-system faults.

The configuration described here is the final intended reference configuration: both controllers run at 8 MHz; USART operates at 9600 baud in asynchronous 8N1 mode; Timer0 supplies custom blocking delays; Timer1 generates the servo waveform; Timer2 supplies the periodic scheduler; ADC3 measures the tamper potentiometer; the analog comparator monitors the divided supply voltage; hardware SPI communicates with a 25LC040A external EEPROM; INT0 wakes the Master from inactivity sleep; and the watchdog uses an approximately two-second timeout.

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Chapter 1

Register Conventions Used in the Firmware

1.1 Read, write, set, clear, and write-one-to-clear operations

A normal read/write control bit is set with a bitwise OR and cleared with a bitwise AND:

```
REGISTER |=  (1U << BIT);   /* set BIT */
REGISTER &= ~(1U << BIT);   /* clear BIT */
```

Some status flags, including `OCF0`, `OCF2`, `INTF0`, `ACI`, and `TXC`, are *write-one-to-clear*. Writing a zero leaves them unchanged; writing a one clears them. The safest form is a direct assignment:

```
TIFR = (1U << OCF0);
GIFR = (1U << INTF0);
ACSR = preserved_control_bits | (1U << ACI);
```

A read-modify-write expression such as `TIFR |= (1U << OCF0)` can unintentionally clear other flags that were already set.

1.2 Atomic sections and SREG preservation

The AVR is an 8-bit processor. Multi-byte variables such as the 32-bit timestamp can be modified by an interrupt while foreground code is reading them. The firmware therefore saves `SREG`, executes `cli()`, accesses the shared object, and restores the saved register. Restoring the complete saved value preserves the interrupt state that existed before the function was called.

Chapter 2

GPIO Registers and Pin Multiplexing

2.1 DDRx, PORTx, and PINx

Each I/O port has three registers. For any pin number n:

Table 2.1: General AVR GPIO bit behavior.

Register bit	Value	Electrical mode	Meaning in this project
`DDRx[n]`	0	Input	Used for keypad columns, RXD, MISO, ADC3, AIN1, and the INT0 button.
`DDRx[n]`	1	Output	Used for LCD buses, keypad rows, servo OC1A, SPI SS/MOSI/SCK, Slave LEDs, and buzzer control.
`PORTx[n]` with DDR=0	0	Input, pull-up disabled	Required for analog inputs and externally driven digital inputs.
`PORTx[n]` with DDR=0	1	Input, internal pull-up enabled	Used for the wake button and keypad columns so an open switch has a defined HIGH state.
`PORTx[n]` with DDR=1	0/1	Output LOW/HIGH	Drives digital signals and output idle levels.
`PINx[n]`	read	Pin sample	Reads keypad columns and other digital input levels.

2.2 Master pin configurations

Table 2.2: Master GPIO configuration by function.

Pin(s)	Register configuration	Idle level	Purpose
PC0–PC7	`DDRC=0xFF`	Last LCD byte	Eight-bit LCD data bus. JTAG must be disabled because PC2–PC5 are shared with JTAG.
PA0	DDRA0=1	LOW for command	LCD RS: LOW selects command register, HIGH selects data register.
PA1	DDRA1=1 or disconnected when R/W is grounded	LOW	LCD write-only mode.
PA2	DDRA2=1	LOW	LCD Enable pulse.

Pin(s)	Register configuration	Idle level	Purpose
PA3/ADC3	DDRA3=0, PORTA3=0	Analog voltage	Tamper potentiometer input; pull-up is disabled to avoid biasing the measurement.
PA4–PA7	output	HIGH	Keypad rows; one row is driven LOW during scanning.
PD3, PD4, PB0, PB1	input with pull-up	HIGH	Keypad columns. A pressed key connected to the selected LOW row is detected as LOW.
PD5/OC1A	output	PWM	Servo control waveform generated by Timer1.
PD2/INT0	input with pull-up	HIGH	Wake button; pressing connects the pin to ground.
PB3/AIN1	input, no pull-up	Analog voltage	Divided monitored supply connected to comparator negative input.
PB4/SS	output	HIGH	25LC040A chip select; HIGH deselects the memory.
PB5/MOSI	output	LOW or last bit	SPI data from Master to EEPROM.
PB6/MISO	input	Driven by EEPROM	SPI data from EEPROM to Master.
PB7/SCK	output	LOW in mode 0	SPI serial clock.
PD0/RXD	peripheral input	HIGH idle	Receives terminal commands.
PD1/TXD	peripheral output	HIGH idle	Sends Slave events and terminal log text.

2.3 Slave pin configurations

The Slave uses PC0–PC7 as the LCD data bus, PA0 as RS, PA2 as Enable, PA3 as the active-buzzer control, PD6 as the green LED, PD7 as the red LED, and PD0/RXD as the USART input. Its LCD R/W pin is tied to ground and therefore is not controlled by a microcontroller register bit.

Chapter 3

CPU Status, JTAG, Reset Flags, Sleep, and External Interrupts

3.1 SREG - AVR Status Register

Table 3.1: `SREG` bits and their project relevance.

Bit	Name	Hardware purpose	Project use
7	I	Global interrupt enable. Cleared on ISR entry and restored by RETI.	Explicitly controlled by `sei()` and `cli()`. The complete register is saved and restored around atomic sections.
6	T	Bit-copy storage for BST/BLD instructions.	Not intentionally used.
5	H	Half-carry arithmetic flag.	Not configured.
4	S	Sign flag, $N \oplus V$.	Not configured.
3	V	Two's-complement overflow flag.	Not configured.
2	N	Negative-result flag.	Used implicitly by compiled arithmetic, not configured directly.
1	Z	Zero-result flag.	Used implicitly by compiled comparisons.
0	C	Carry flag.	Used implicitly by compiled arithmetic.

3.2 MCUCSR - MCU Control and Status Register

Table 3.2: `MCUCSR` bits.

Bit	Name	Purpose	Project configuration
7	JTD	Disables the JTAG interface when written as required by the timed double-write sequence.	Written to one twice during startup so PORTC can be used by the LCD.
6	ISC2	Selects INT2 rising or falling edge.	Not used; INT2 is not part of the design.

Bit	Name	Purpose	Project configuration
5	-	Reserved.	Left zero.
4	JTRF	JTAG reset flag.	Not used.
3	WDRF	Watchdog reset flag.	Read at the beginning of Master startup, then cleared. A set flag causes `WDT_RESET` to be reported and logged.
2	BORF	Brown-out reset flag.	Not used by firmware; brown-out detection is still recommended for physical hardware.
1	EXTRF	External reset flag.	Not used in the application logic.
0	PORF	Power-on reset flag.	Not used in the application logic.

3.3 MCUCR - sleep and INT0 sense control

Table 3.3: `MCUCR` bits used by sleep and INT0.

Bit	Name	Purpose	Project value
7	SE	Enables the SLEEP instruction.	Set immediately before `sleep_cpu()` and cleared after wake-up by avr-libc sleep macros.
6:4	SM2:SM0	Sleep-mode selection.	000 for Idle during power-failure restoration wait; 010 for Power-down during inactivity sleep.
3:2	ISC11:ISC10	INT1 sense control.	Not used.
1:0	ISC01:ISC00	INT0 sense control.	00 selects low-level wake. This is suitable for a button that pulls PD2 to ground and can wake from Power-down.

3.4 GICR and GIFR

Table 3.4: External-interrupt control and flag bits.

Register	Bit	Meaning	Project use
GICR	INT0 (bit 6)	Enables external interrupt 0.	Enabled only immediately before inactivity Power-down; disabled in the ISR and after wake.
GICR	INT1, INT2	Enable other external interrupts.	Left zero.
GICR	IVSEL, IVCE	Relocate interrupt vectors through a timed sequence.	Left zero; vectors remain in the application section.
GIFR	INTF0 (bit 6)	INT0 interrupt flag, write-one-to-clear.	Cleared before enabling INT0 and after waking to remove stale requests.
GIFR	INTF1, INTF2	Flags for INT1 and INT2.	Not used.

Chapter 4

Watchdog Timer Register

4.1 WDTCR - Watchdog Timer Control Register

Table 4.1: `WDTCR` bits.

Bit	Name	Purpose	Project configuration
7:5	-	Reserved.	Zero.
4	WDTOE	Opens the four-cycle window required to clear WDE.	Used internally by `wdt_disable()`.
3	WDE	Enables the watchdog reset function.	Enabled during normal runtime and disabled before deliberate sleep/standby.
2:0	WDP2:WDP0	Watchdog prescaler.	111 through `WDTO_2S`, giving approximately 2.1 s at 5 V.

The avr-libc watchdog macros implement the required timed write sequence. The main loops call `wdt_reset()` repeatedly. Long log dumps also reset the watchdog inside the record loop.

Chapter 5

Timer0 - Polled Blocking Delays

5.1 TCCR0 bit configuration

Table 5.1: `TCCR0` bits used by the delay routines.

Bit	Name	Purpose	Project setting
7	FOC0	Forces an output compare in non-PWM modes.	Zero; no OC0 pin action is required.
6	WGM00	Waveform generation bit 0.	Zero. Together with WGM01=1, selects CTC mode.
5:4	COM01:COM00	OC0 pin behavior.	00; OC0 is disconnected.
3	WGM01	Waveform generation bit 1.	One during delays, selecting CTC mode.
2:0	CS02:CS00	Timer0 clock selection.	011 for `delay_ms()` (CPU/64); 001 for corrected `delay_us()` (CPU/1); 000 when stopped.

5.2 TCNT0, OCR0, TIMSK, and TIFR

Table 5.2: Other Timer0 registers.

Register	Bit/value	Meaning	Project use
TCNT0	0	Current 8-bit counter.	Reset before each requested delay unit.
OCR0	124	Compare value.	With CPU/64, 125 counts produce exactly 1 ms at 8 MHz.
OCR0	7	Compare value.	With no prescaler, eight counts produce a nominal 1 us at 8 MHz.
TIMSK	OCIE0=0	Timer0 compare interrupt enable.	Cleared because the code polls the flag instead of executing an ISR.
TIFR	OCF0	Timer0 compare flag.	Polled until set and cleared by writing one.

For the millisecond delay,

$$T = (124 + 1)\frac{64}{8\,000\,000} = 1 \text{ ms.}$$

For the corrected microsecond delay,

$$T = (7 + 1)\frac{1}{8\,000\,000} = 1 \text{ us.}$$

The routines save `TCCR0`, `TIMSK`, `OCR0`, and `TCNT0`, stop the timer, perform the delay, clear the final flag, and restore the saved configuration with the clock-selection register restored last.

Chapter 6

Timer1 - Servo PWM

6.1 TCCR1A

Table 6.1: `TCCR1A` bits.

Bit	Name	Purpose	Project setting
7	COM1A1	OC1A compare output control.	One. With COM1A0=0, produces non-inverting PWM on PD5/OC1A.
6	COM1A0	OC1A compare output control.	Zero.
5:4	COM1B1:COM1B0	OC1B behavior.	00; channel B is unused.
3	FOC1A	Force channel-A compare in non-PWM mode.	Zero in PWM mode.
2	FOC1B	Force channel-B compare.	Zero.
1	WGM11	Waveform generation bit.	One as part of mode 14.
0	WGM10	Waveform generation bit.	Zero as part of mode 14.

6.2 TCCR1B

Table 6.2: `TCCR1B` bits.

Bit	Name	Purpose	Project setting
7	ICNC1	Input-capture noise canceler.	Zero; input capture is unused.
6	ICES1	Input-capture edge select.	Zero; input capture is unused.
5	-	Reserved.	Zero.
4	WGM13	Waveform generation bit.	One.
3	WGM12	Waveform generation bit.	One.
2	CS12	Clock select.	Zero.
1	CS11	Clock select.	One, selecting CPU/8.
0	CS10	Clock select.	Zero.

The complete mode value is $WGM13:0 = 1110$, Fast PWM with `ICR1` as TOP. The timer tick is 1 us. `ICR1`=19999 creates 20,000 counts and therefore a 20 ms, 50 Hz frame. `OCR1A`=1000 produces a 1 ms locked pulse, and `OCR1A`=1500 produces a 1.5 ms 90-degree unlocked pulse.

Chapter 7

Timer2 - Periodic Scheduler

7.1 TCCR2

Table 7.1: `TCCR2` bits.

Bit	Name	Purpose	Project setting
7	FOC2	Force output compare in non-PWM modes.	Zero.
6	WGM20	Waveform generation bit 0.	Zero.
5:4	COM21:COM20	OC2 pin behavior.	00; OC2 is disconnected.
3	WGM21	Waveform generation bit 1.	One, selecting CTC mode.
2:0	CS22:CS20	Clock select.	111, selecting CPU/1024.

7.2 Timer2 compare and interrupt registers

Table 7.2: Timer2 count, compare, mask, and flag settings.

Register	Bit/value	Meaning	Project use
TCNT2	0	Current counter value.	Reset during initialization.
OCR2	77	Compare value.	78 counts at 128 us per count produce 9.984 ms.
TIMSK	OCIE2=1	Timer2 compare interrupt enable.	Enables `TIMER2_COMP_vect`.
TIMSK	TOIE2=0	Timer2 overflow interrupt enable.	Not required in CTC mode.
TIFR	OCF2	Compare flag, write-one-to-clear.	Cleared before enabling the interrupt.

Ten compare interrupts produce a 99.84 ms service flag. Ten service intervals produce a 998.4 ms software second. The resulting -0.16 percent timing error is acceptable for human-scale five-, ten-, and thirty-second intervals.

Chapter 8

USART Registers

8.1 Baud-rate registers UBRRH and UBRRL

The normal asynchronous formula is

$$UBRR = \frac{F_{CPU}}{16\,BAUD} - 1.$$

For 8 MHz and 9600 baud, $UBRR = 51.083$, rounded to 51. The actual rate is 9615.38 baud, an error of approximately +0.16 percent.

`UBRRL` stores the lower eight divisor bits. The low bits of `UBRRH` store the remaining divisor bits. On the ATmega32, `UBRRH` shares an I/O address with `UCSRC`; `URSEL` must be zero when writing `UBRRH` and one when writing `UCSRC`.

8.2 UCSRA - USART Control and Status Register A

Table 8.1: `UCSRA` bits.

Bit	Name	Purpose	Project use
7	RXC	Receive complete flag.	Polled by `usart_rx_available()` and `usart_rx()`.
6	TXC	Transmission complete flag.	Cleared by writing one and polled before power-failure standby so the final stop bit has left TXD.
5	UDRE	Data register empty flag.	Polled before writing each byte to `UDR`.
4	FE	Frame error.	Not explicitly handled.
3	DOR	Data overrun.	Not explicitly handled.
2	PE	Parity error.	Not used because parity is disabled.
1	U2X	Double asynchronous speed.	Zero; normal speed is used.
0	MPCM	Multi-processor communication mode.	Zero.

8.3 UCSRB

Table 8.2: `UCSRB` bits.

Bit	Name	Purpose	Project use
7	RXCIE	Receive-complete interrupt enable.	Zero; both units poll the receiver.
6	TXCIE	Transmit-complete interrupt enable.	Zero.
5	UDRIE	Data-register-empty interrupt enable.	Zero; transmission is blocking and polled.
4	RXEN	Receiver enable.	One on Master and Slave.
3	TXEN	Transmitter enable.	One on Master; zero on receive-only Slave.
2	UCSZ2	Character size bit 2.	Zero; combined with UCSZ1:0=11 to select 8-bit characters.
1	RXB8	Ninth received bit.	Not used.
0	TXB8	Ninth transmitted bit.	Not used.

8.4 UCSRC

Table 8.3: `UCSRC` bits.

Bit	Name	Purpose	Project setting
7	URSEL	Selects UCSRC rather than UBRRH at the shared address.	One whenever UCSRC is written.
6	UMSEL	Synchronous/asynchronous selection.	Zero for asynchronous mode.
5:4	UPM1:UPM0	Parity mode.	00, parity disabled.
3	USBS	Stop-bit selection.	Zero, one stop bit.
2:1	UCSZ1:UCSZ0	Character size.	11, with UCSZ2=0, giving eight data bits.
0	UCPOL	Clock polarity in synchronous mode.	Zero and irrelevant in asynchronous mode.

`UDR` is the transmit/receive data register. Writing it starts transmission when the transmitter is enabled; reading it retrieves the oldest received byte and clears RXC when no additional byte remains buffered.

Chapter 9

Hardware SPI Registers

9.1 SPCR - SPI Control Register

Table 9.1: SPCR bits.

Bit	Name	Purpose	Project setting
7	SPIE	SPI interrupt enable.	Zero; transfers are polled.
6	SPE	SPI peripheral enable.	One during runtime; cleared during power-failure standby.
5	DORD	Data order.	Zero, most-significant bit first, matching the 25LC040A.
4	MSTR	Master/slave selection.	One; the ATmega32 generates SCK.
3	CPOL	Clock polarity.	Zero, idle clock LOW.
2	CPHA	Clock phase.	Zero, data sampled on the leading rising edge. This is SPI mode 0.
1:0	SPR1:SPR0	SPI clock prescaler.	01, selecting CPU/16 when SPI2X=0.

9.2 SPSR and SPDR

Table 9.2: SPSR and SPDR.

Register/bit	Name	Purpose	Project use
SPSR bit 7	SPIF	SPI transfer-complete flag.	Polled after writing SPDR.
SPSR bit 6	WCOL	Write-collision flag.	Not expected because the driver waits for each transfer.
SPSR bit 0	SPI2X	Doubles SPI speed.	Zero. Combined with SPR1:0=01, SCK is 8 MHz/16 = 500 kHz.
SPDR	Data register	Starts a transfer when written and contains the received byte when complete.	Used for command, address, status, and data bytes.

Chapter 10

ADC Registers

10.1 ADMUX

Table 10.1: `ADMUX` bits.

Bit	Name	Purpose	Project setting
7:6	REFS1:REFS0	ADC voltage reference.	01, AVCC reference with a capacitor on AREF.
5	ADLAR	Left-adjust result.	Zero, right-adjusted 10-bit result.
4:0	MUX4:MUX0	Input channel/-gain selection.	00011 for single-ended ADC3. The read function preserves reference and adjustment bits while changing the channel.

10.2 ADCSRA

Table 10.2: `ADCSRA` bits.

Bit	Name	Purpose	Project setting
7	ADEN	ADC enable.	One during runtime; cleared in power-failure standby.
6	ADSC	Start conversion and conversion-busy indication.	Written one for each sample and polled until hardware clears it.
5	ADATE	Auto-trigger enable.	Zero; conversions are software-triggered.
4	ADIF	Conversion-complete flag, write-one-to-clear.	Not used because ADSC is polled.
3	ADIE	ADC interrupt enable.	Zero.
2:0	ADPS2:ADPS0	ADC prescaler.	110, divide by 64, producing a 125 kHz ADC clock.

With a 5 V reference, one ADC count represents approximately $5/1024 = 4.883$ mV. A threshold of 50 counts therefore corresponds to approximately 244 mV. A normal conversion takes 13 ADC clock cycles, or approximately 104 us at 125 kHz; the first conversion after enabling the ADC takes 25 cycles.

10.3 ADCL and ADCH

Because ADLAR=0, the result is right-adjusted: ADCL contains bits ADC7:0 and the two low positions of ADCH contain ADC9:8. The avr-libc `ADC` macro reads the combined 16-bit register in the correct order.

Chapter 11

Analog Comparator and Power Monitoring

11.1 ACSR - Analog Comparator Control and Status Register

Table 11.1: `ACSR` bits.

Bit	Name	Purpose	Project setting
7	ACD	Analog comparator disable.	Zero; comparator remains enabled for monitoring and standby wake.
6	ACBG	Bandgap select for comparator positive input.	One; internal bandgap replaces AIN0 as the positive input.
5	ACO	Live comparator output.	Read-only. One means bandgap voltage is greater than the divided voltage on AIN1, interpreted as power failure.
4	ACI	Analog comparator interrupt flag, write-one-to-clear.	Cleared before enabling or re-enabling comparator interrupts.
3	ACIE	Comparator interrupt enable.	One during monitoring and restoration standby; zero during ordinary inactivity Power-down.
2	ACIC	Connect comparator to Timer1 input capture.	Zero.
1:0	ACIS1:ACIS0	Comparator interrupt mode.	00, interrupt on output toggle. Both failure and restoration transitions are therefore detected.

11.2 SFIOR ACME bit

`ACME` in `SFIOR` controls whether the ADC multiplexer can replace AIN1 as the negative comparator input. The project clears ACME so the dedicated AIN1/PB3 pin is used. This avoids unintended coupling between ADC channel selection and the power monitor.

11.3 Software confirmation state

The comparator ISR does not directly report a failure. It records the candidate ACO state and loads a 10-tick counter. Timer2 decrements that counter every 9.984 ms. When it reaches zero, foreground code compares the live ACO state with the stored candidate and accepts the transition only when they still agree. This produces approximately 100 ms of software filtering without performing slow logging or USART operations inside the comparator ISR.

Chapter 12

Internal EEPROM Registers

12.1 EEARH and EEARL

The ATmega32A contains 1024 EEPROM bytes, requiring ten address bits. `EEARL` contains address bits 7:0, and the low two bits of `EEARH` contain address bits 9:8. The C header exposes the combined register as `EEAR`.

12.2 EEDR and EECR

Table 12.1: Internal EEPROM control bits.

Bit	Name	Purpose	Project use
EECR bit 3	EERIE	EEPROM-ready interrupt enable.	Zero; the firmware polls.
EECR bit 2	EEMWE	Master write authorization.	Set immediately before EEWE. Hardware clears it after four CPU cycles.
EECR bit 1	EEWE	Starts writing and remains one while programming is active.	Polled before every read/write; set within four cycles after EEMWE.
EECR bit 0	EERE	Starts an EEPROM read.	Set after loading EEAR; the selected byte appears in EEDR.
EEDR	-	EEPROM data register.	Holds the byte being written or the byte just read.

The EEMWE-to-EEWE sequence is protected with `cli()` and restoration of the saved SREG value. The internal EEPROM stores a signature byte, four password characters, and the failed-attempt counter.

Chapter 13

25LC040A Instruction and Status Configuration

13.1 Physical pins

The external memory uses CS, SO, WP, VSS, SI, SCK, HOLD, and VCC. WP and HOLD are tied HIGH because hardware write protection and bus hold are not used. CS is controlled by PB4 and remains HIGH when the device is idle.

13.2 Instruction opcodes

Table 13.1: 25LC040A instructions.

Instruction	Opcode	Used?	Purpose
READ	0x03	Yes	Reads one or more bytes beginning at the selected address. For A8=1, bit 3 is set and the transmitted instruction becomes 0x0B.
WRITE	0x02	Yes	Writes data beginning at the selected address. For A8=1, the transmitted instruction becomes 0x0A.
WREN	0x06	Yes	Sets the volatile Write Enable Latch before every write.
WRDI	0x04	No	Clears the Write Enable Latch. A completed WRITE also clears it automatically.
RDSR	0x05	Yes	Reads the status register so WIP can be polled.
WRSR	0x01	No	Writes the nonvolatile block-protection bits.

13.3 Ninth address bit

The 25LC040A contains 512 bytes, so the address range is 0x000–0x1FF and nine address bits are needed. Only one separate address byte is transmitted. Address bit A8 is inserted into bit 3 of the READ or WRITE instruction byte:

```
command = base_opcode | ((address & 0x0100U) ? 0x08U : 0x00U);
spi_transfer(command);
spi_transfer((uint8_t)(address & 0xFFU));
```

Thus address 0x055 uses READ 0x03 followed by 0x55, while address 0x155 uses READ 0x0B followed by 0x55.

13.4 STATUS register bits

Table 13.2: 25LC040A STATUS register.

Bit	Name	Meaning	Project use
7:4	X	Unimplemented/reserved in this device.	Ignored.
3	BP1	Block-protection bit 1.	Left zero; no array block is intentionally protected.
2	BP0	Block-protection bit 0.	Left zero.
1	WEL	Write Enable Latch status, read-only.	Set by WREN. The driver does not separately poll it because it controls the full sequence.
0	WIP	Write In Process, read-only.	Polled after every byte write until it becomes zero.

Chapter 14

LCD Command Bytes Used by the Firmware

Although the LCD command register is external to the ATmega32A, its command bytes are part of the low-level configuration:

Table 14.1: HD44780-compatible LCD command values.

Value	Command	Project meaning
0x38	Function set	8-bit interface, two display lines, standard character font.
0x0C	Display control	Display ON, cursor OFF, blink OFF.
0x06	Entry mode set	Increment DDRAM address after each character; no display shift.
0x01	Clear display	Clears DDRAM and returns cursor home; requires the longest execution delay.
0x08	Display control	Display OFF while retaining DDRAM contents.
0x80+column	Set row-0 address	Positions cursor on the first line.
0xC0+column	Set row-1 address	Positions cursor on the second line.

Chapter 15

Exact Final Register Values at Initialization

Table 15.1: Initialization-value summary.

Subsystem	Register/value	Principal set bits	Reason
Master USART	UBRR=51	-	9600 baud at 8 MHz, normal asynchronous speed.
Master USART	UCSRB=0x18	RXEN, TXEN	Enable receiver and transmitter.
Slave USART	UCSRB=0x10	RXEN	Receive-only remote endpoint.
Both USARTs	UCSRC=0x86	URSEL, UCSZ1, UCSZ0	Write UCSRC and select asynchronous 8N1.
Timer0 ms	TCCR0=0x0B	WGM01, CS01, CS00	CTC, CPU/64.
Timer0 us	TCCR0=0x09	WGM01, CS00	CTC, CPU/1.
Timer1	TCCR1A=0x82	COM1A1, WGM11	Non-inverting OC1A, mode 14.
Timer1	TCCR1B=0x1A	WGM13, WGM12, CS11	Mode 14, CPU/8.
Timer1	ICR1=19999	-	20 ms PWM period.
Timer1	OCR1A=1000 or 1500	-	Locked or unlocked pulse width.
Timer2	TCCR2=0x0F	WGM21, CS22, CS21, CS20	CTC, CPU/1024.
Timer2	OCR2=77	-	9.984 ms compare interval.
Timer2	TIMSK.OCIE2=1	OCIE2	Enable scheduler ISR.
ADC	ADMUX=0x40 then channel 3	REFS0, MUX1, MUX0	AVCC reference, right-adjusted ADC3.
ADC	ADCSRA=0x86	ADEN, ADPS2, ADPS1	Enable ADC, CPU/64.
SPI	SPCR=0x51	SPE, MSTR, SPR0	Enabled master, mode 0, CPU/16.
SPI	SPSR=0x00	-	No double-speed mode.
Comparator	ACSR normal=0x48	ACBG, ACIE	Bandgap positive input, toggle interrupt.
Comparator flag clear	ACSR write includes ACI	ACI	Clear stale comparator flag before enabling.
INT0 sleep	MCUCR ISC01:0=00	-	Low-level wake from Power-down.
INT0 sleep	GICR.INT0=1	INT0	Enable only immediately before inactivity sleep.
Watchdog	WDE=1, WDP2:0=111	WDE, WDP2, WDP1, WDP0	Approximate 2.1 s reset timeout.

Chapter 16

Register Ownership by Source Module

Table 16.1: Source modules and the registers they own or access.

Module	Registers
`main.c`	MCUCSR, MCUCR, GICR, GIFR, TIMSK, TCCR0, TCCR1A, TCCR1B, TCCR2, ADCSRA, SPCR, UCSRB, SREG, GPIO registers, watchdog and sleep macros.
`timer.c`	TCCR0, TCNT0, OCR0, TCCR2, TCNT2, OCR2, TIMSK, TIFR, SREG.
`servo.c`	DDRD, TCCR1A, TCCR1B, ICR1, OCR1A.
`usart.c`	UBRRH, UBRRL, UCSRA, UCSRB, UCSRC, UDR.
`adc.c`	DDRA, PORTA, ADMUX, ADCSRA, ADCL/ADCH through ADC.
`power.c`	DDRB, PORTB, ACSR, SFIOR, SREG.
`spi_eeprom.c`	DDRB, PORTB, SPCR, SPSR, SPDR.
`password.c`	EECR, EEARH/EEARL through EEAR, EEDR, SREG.
`keypad.c`	DDRA, PORTA, PINA, DDRB, PORTB, PINB, DDRD, PORTD, PIND.
`LCD.c`	DDRC, PORTC, DDRA, PORTA, Timer0 delay services.
Slave output logic	DDRA, PORTA, DDRD, PORTD, Timer2 counters.

Chapter 17

Common Register-Level Errors and Their Consequences

Table 17.1: Common faults prevented by the final configuration.

Error	Result
Proteus clock set to 1 MHz while firmware assumes 8 MHz	USART divisor, timer delays, and servo period become eight times wrong.
JTAG not disabled	PC2–PC5 do not behave as a normal LCD data bus.
ADC pull-up enabled	Tamper voltage is biased toward VCC and threshold behavior becomes incorrect.
Keypad pull-ups disabled	Open columns float and false key presses appear.
Writing zero to OCF0, OCF2, INTF0, ACI, or TXC	The stale flag remains set.
Using read-modify-write on a shared write-one-to-clear flag register	Other pending flags may be cleared accidentally.
Starting EEPROM write without EEMWE-to-EEWE timed sequence	The internal EEPROM byte is not programmed.
Failing to issue WREN before 25LC040A WRITE	External EEPROM ignores the write.
Ignoring WIP after external EEPROM WRITE	A following command may arrive while the device is still busy.
Setting CPOL or CPHA incorrectly	EEPROM command and data bits are sampled on the wrong edges.
Using OCR0=0 for the simulated microsecond delay	Compare behavior can be unreliable in Proteus and may stall LCD initialization.
Reading the comparator immediately after enabling bandgap	Startup state may be wrong, especially when RESET occurs during an existing low-voltage condition.
Enabling USART TX on the Slave TXD line connected to another transmitter	Bus contention can occur.

Chapter 18

Conclusion

The SmartSafe firmware uses the ATmega32A register set in a deliberately partitioned manner. Timer1 belongs to servo PWM, Timer2 belongs to application scheduling, and Timer0 is borrowed only for polled delays. USART and SPI are configured as simple polled interfaces because the traffic volume is low. ADC3 is configured as a high-impedance analog input, while keypad and wake-button inputs use internal pull-ups. The analog comparator remains available as the power-restoration wake source even after most other peripherals are disabled. Internal and external EEPROM operations use different register and protocol mechanisms, but both require strict write sequencing and busy-state handling.

Understanding the individual bits is essential because the most difficult faults in this project were not high-level algorithm errors; they were register-level timing, shared-pin, flag-clearing, and peripheral-state issues. The tables in this document provide a direct trace from every significant line of initialization code to the corresponding hardware behavior.

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